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CODEX OF REST

*The architecture of a night, the debt a short week leaves
behind, and the difference between resting and merely
switching off*

VILLUMINATIONS

Fine edition, 2026

Original work · First edition, 2026
villuminations.com

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How to read this book. It is a book about *sleep hygiene* for healthy people. Persistent insomnia, apnoea and excessive daytime sleepiness are clinical conditions: nothing that follows diagnoses or treats them, and it does not replace medical care. If you have been sleeping badly for more than three months, or if someone has told you that you stop breathing in your sleep, that is grounds for an appointment, not for a chapter.

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CHAPTER 1

THE HALF OF THE DAY NOBODY TRAINS

An adult who follows the activity guidelines spends about four hours a week training. That same adult spends between forty and fifty hours a week asleep, or trying to be. It is by a wide margin the activity they give most of their life to, and the only one almost nobody counts as part of their physical preparation.

WHAT THIS BOOK CLAIMS

Sleep is not what is left over from the day: it is the phase in which almost everything that training and food merely set up actually happens. And unlike motivation, it has a structure that is known, measurable and fairly easy to manipulate with simple decisions.

THE RECOMMENDATION, AND WHERE IT COMES FROM

ADULT	7 hours or more a night, on a regular basis
CYCLE	about 90 minutes
CYCLES PER NIGHT	4 to 6
THE ARITHMETIC	6 cycles of 90 minutes make 9 hours; 4 cycles make 6. The 7-hour recommendation falls exactly inside that range, and not by chance.

WHAT THIS BOOK DOES NOT CLAIM

- It does not treat sleep disorders. Apnoea, chronic insomnia, restless legs syndrome and narcolepsy are medical diagnoses with treatments, and none of them is fixed with sleep hygiene. The final chapter says how to recognise that it is time to see someone.
- It does not promise you will sleep better tonight. Almost everything here works over weeks, and chasing the result of one particular night is one of the things that works worst.
- It does not give an optimal bedtime. There is real individual variation in chronotype, and forcing someone with an evening tendency to get up early, just like that, does not turn them into a morning person.
- It does not cover supplements or medication. That is the territory of whoever writes your prescription, and melatonin in particular is a case where the dose, the timing and the reason matter far more than over-the-counter availability suggests.

WHAT IS WELL ESTABLISHED AND WORTH KEEPING IN MIND

Sustained sleep deprivation is associated with worse cognitive performance, worse glucose control, greater appetite, higher accident risk and poorer recovery from training. It is one of the most consistent relationships in all of public health, and it is why an institution as unsentimental as an army treats sleep as ammunition rather than as weakness.

CHAPTER 2

THE INTERNAL CLOCK

Behind sleep there is a clock that does not measure wall-clock hours but a cycle of its own, a little longer than a day, and that is reset every morning by light. Almost all the sleep problems of healthy people are, in reality, problems with that clock.



THE BIOLOGICAL DAY, MILESTONE BY MILESTONE The hours are counted from your usual waking time and correspond to an intermediate chronotype. None of these milestones is decided at will: they are dragged along by changing your get-up time and your light exposure.

THE MILESTONES OF THE DAY

-2 H · CORE TEMPERATURE MINIMUM	The lowest point of the day. Waking here is what feels worst
+0 H · WAKING	Cortisol rises sharply: this is normal and it is useful
+0.6 H · MORNING CORTISOL PEAK	Thirty to forty-five minutes after opening your eyes
+7 H · AFTERNOON DIP IN ALERTNESS	Food does not cause it: it happens fasted too. It is the slot for the short nap
+10 H · PEAK STRENGTH AND TEMPERATURE	The window of best performance on almost every physical test
+14 H · WAKE MAINTENANCE ZONE	The two or three hours before sleep when falling asleep is hardest, even when tired
+15 H · MELATONIN ONSET	Under dim light. Bright light at this hour delays it
+16 H · SLEEP	If the schedule is stable

THE THREE YOU CAN DO MOST WITH

The core temperature minimum falls about two hours before your usual waking time, and it is the worst possible moment to get up. Someone who moves the alarm two hours earlier on a one-off day is not stealing two ordinary hours of sleep: they are waking at the bottom of the well, and hence the disproportionate feeling.

The wake maintenance zone is the most reassuring milestone to know about. The two or three hours before your usual sleep onset are, paradoxically, among the hardest of the day for falling asleep: the clock pushes upwards just before it lets go. That is why going to bed two

hours earlier than usual the night before an early start almost never works, and why it does not mean you have insomnia.

And melatonin onset, which happens under dim light a couple of hours before sleep, is what explains the last hour without bright screens and without the ceiling light on. Not because a screen poisons you: because bright light at that hour delays the onset, and the delay is paid for the next morning.



CHRONOTYPE, AND WHY IT IS NOT AN EXCUSE

There is real individual variation in time-of-day preference and it has a biological basis: some people's clock naturally runs early and others' runs late. It is not a question of discipline, and mocking it makes about as much sense as mocking someone's height.

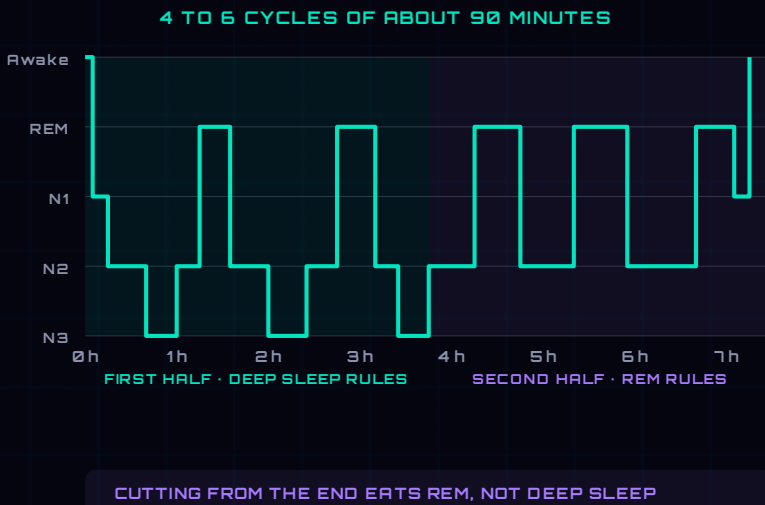
What is true is that the clock can be shifted within a margin, and that the tool for doing it is light: bright light shortly after waking advances the clock; bright light at night delays it. Anyone who wants to get up earlier has their lever there, and not in going to bed sooner to stare at the ceiling.

WHEN THE OFFSET STOPS BEING A CHRONOTYPE

If the time-of-day preference is so extreme that it prevents sleeping on socially workable hours for months on end, with serious consequences at work or in study, that has a clinical name and a treatment, which includes light protocols at specific times. It is not fixed with willpower and it deserves an appointment.

THE ARCHITECTURE OF A NIGHT

Sleeping is not one state: it is four, alternating in an order that is not random and that repeats in cycles. That order is the most useful piece of information in this book, because it explains exactly what is lost when you sleep too little.



THE COURSE OF A FULL NIGHT From 4 to 6 cycles of about 90 minutes. Deep sleep is concentrated in the first half of the night, and the stretches of rapid eye movement sleep lengthen with every cycle, dominating the second half. The night is not symmetrical, and that asymmetry is the key to the whole chapter.

THE FOUR STAGES

N1	TRANSITION Very light. You may deny having slept at all	1 to 5 min
N2	LIGHT SLEEP Much of procedural memory is consolidated here	Most of the night
N3	DEEP SLEEP The one for physical recovery. If sleep is cut from the end, this one survives	Concentrated in the first half
REM	RAPID EYE MOVEMENT Concentrated in the second half. If sleep is cut from the end, this is the one you	It grows with every cycle

SLEEP ARCHITECTURE · FOUR STATES, NOT ONE

WHAT EACH STAGE DOES The first two states are light sleep; the third is deep sleep, associated with physical recovery; the fourth is rapid eye movement sleep. A brief awakening between cycles is normal and does not mean anything is wrong.

IN A TABLE

N1 · TRANSITION	1 to 5 min · Very light. You may deny having slept at all
N2 · LIGHT SLEEP	Most of the night · Much of procedural memory is consolidated here
N3 · DEEP SLEEP	Concentrated in the first half · The one for physical recovery. If sleep is cut from the end, this one survives
REM · RAPID EYE MOVEMENT	It grows with every cycle · Concentrated in the second half. If sleep is cut from the end, this is the one you lose



THE CONSEQUENCE ALMOST NOBODY HAS HEARD

If the night were homogeneous, cutting an hour would cost the same from either end. It is not. Because deep sleep comes first and rapid eye movement sleep comes later, removing the last hour does not remove a little of each: it removes almost exclusively the second, which is precisely the one that grows most in the final cycles.

That means someone sleeping five hours keeps most of their deep sleep and loses an enormous fraction of the other. And it explains something many people have noticed without knowing why: after a short night you can feel physically acceptable and be clearly worse in mood, concentration and memory.

It works the other way round too, and this matters more for anyone who trains: going to bed late and getting up at the same time eats into the second half's sleep; going to bed on time and getting up earlier does the same. Both of the usual ways of cutting attack the same side of the night.

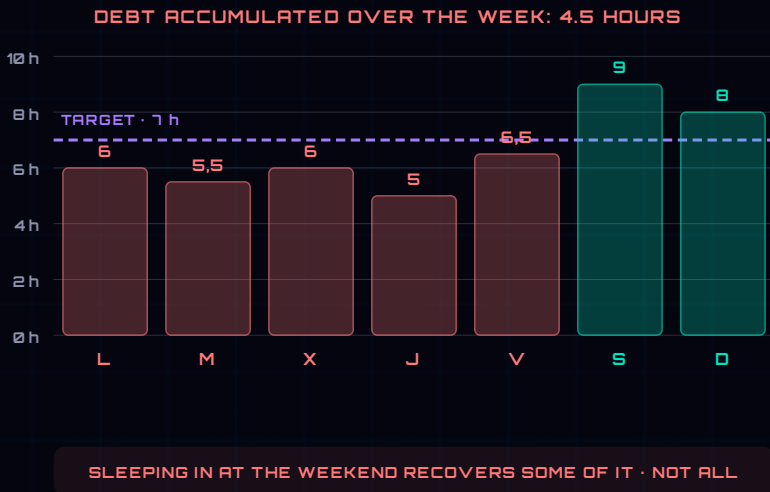
WHY THE MYTH OF WAKING BETWEEN CYCLES ONLY HALF WORKS

It is true that waking in the middle of a deep stretch produces that thick, groggy feeling, and that waking in light sleep is more pleasant. What is not true is that the apps claiming to work out your cycle from a phone on the bedside table can do so accurately: cycles do not last exactly ninety minutes, nor are they equal to one another or between people. The reliable way to wake in light sleep is to sleep enough, because the last cycles are almost entirely light sleep and REM.

CHAPTER 4

THE DEBT

The phrase “sleep debt” sounds like a metaphor and describes what happens rather well: the deficit accumulates, produces measurable effects before you notice them, and is not settled in one go.



AN ORDINARY WEEK AND WHAT IT LEAVES BEHIND Five nights below the 7-hour target and two of recovery at the weekend. The week's debt is not cancelled: sleeping in on Saturday recovers part of the deficit—above all the deep sleep, which has priority—but not all of it, and in exchange it shifts the schedule and makes Monday harder.

THE THREE FACTS THAT MAKE THE DEBT DANGEROUS

IT ACCUMULATES

Several nights of six hours produce a deterioration comparable to whole nights without sleep, reached little by little.

IT IS NOT PERCEIVED

The subjective sense of sleepiness flattens out while performance keeps falling. Someone who has slept badly for a week believes they have adapted, and what they have actually done is lose the ability to notice.

IT IS REPAID WITH A SURCHARGE

The body gives priority to recovering deep sleep, and on recovery nights more of it appears than usual. The sleep of the second half recovers less well — and that is the one that had been lost.

The second point is the one with the most consequences outside the gym. Feeling fine is not a reliable indicator of being fine, and the situation where that really matters is behind the wheel. Driving with accumulated sleep debt degrades attention and reaction time in documented ways, and unlike other causes of poor driving, this one is not compensated for by willpower.



THE WEEKEND: WHAT IT RECOVERS AND WHAT IT SPOILS

Sleeping more on Saturday and Sunday is better than not doing it, and that is worth saying because the advice to keep rigid hours sounds

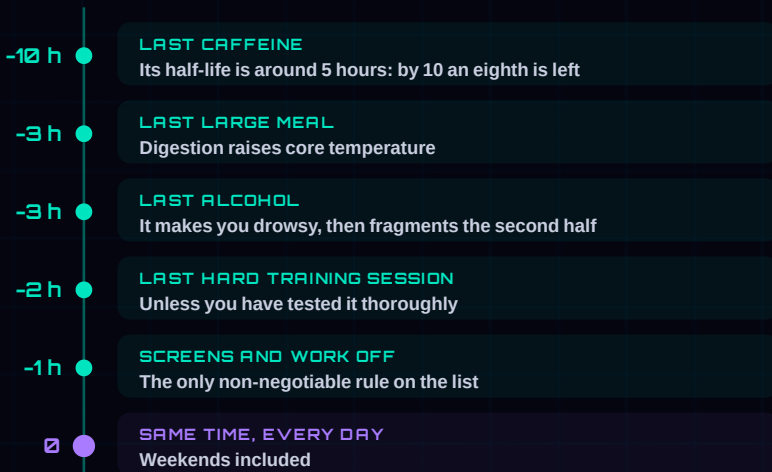
like sleeping in is wrong. It is not. What happens is that it has a cost: getting up three hours later on Sunday is, for the internal clock, much like travelling to another time zone, and on Sunday night sleep arrives late. It is the most common cause of Sunday insomnia and the dreadful Monday.

- Recover, yes, but within bounds: up to an hour more than usual barely moves the clock; three hours move it considerably.
- It is better to move bedtime earlier at the weekend than to move the get-up time later. You get the same hours without shifting the anchor.
- Get-up time is the strongest anchor of the internal clock, above all when it comes with light. If you can fix only one thing about your schedule, fix that one.

CHAPTER 5

THE COUNTDOWN

Sleep hygiene is usually presented as a disordered list of tips of uneven effectiveness. Ordered in time it works much better, because every item has an hour after which it can no longer be fixed.



COUNTDOWN FROM BEDTIME

WHAT GETS CUT AND WHEN, COUNTING BACK FROM BEDTIME The times are reasonable orders of magnitude, not exact thresholds. The caffeine one comes from its half-life, which is around five hours in an average adult: at ten hours roughly an eighth of the dose is still circulating.

CAFFEINE, WITH ITS ARITHMETIC

The half-life is the time the body takes to clear half of it. With a half-life of about five hours, a coffee at five in the afternoon leaves half its caffeine at ten at night and a quarter at three in the morning. It does not necessarily stop you falling asleep, and it does reduce the amount of deep sleep that night — an effect you do not perceive.

There is also a large individual variation in clearance speed: some people metabolise caffeine twice as fast as others. That is why the argument “it does not affect me” can be literally true for some people and a perception error for others. The way to find out is to drop it for two weeks and compare, not to reason about it.

ALCOHOL, WHICH IS THE MOST WIDESPREAD DECEPTION

Alcohol shortens the time it takes to fall asleep, and that is where its reputation for helping you sleep comes from. What it does afterwards is the opposite: it fragments the second half of the night and suppresses part of the rapid eye movement sleep, which we have already seen is precisely the sleep that lives there. The result is a night that starts easy and ends badly, with more awakenings and a worse sense of rest.



LIGHT AND TEMPERATURE, WHICH ARE THE TWO BIG LEVERS

THE TWO SIGNALS THE INTERNAL CLOCK ACTUALLY READS

LIGHT IN THE MORNING	It is the most powerful signal for setting the clock. Ten or fifteen minutes of outdoor light shortly after getting up do more than any night-time ritual, and it is the most underrated piece of advice in the chapter.
DARKNESS AT NIGHT	Dim, warm light in the last hour. The problem with screens is not only the light: it is what you do on them, which is activating.
TEMPERATURE	Sleeping is a process that requires core temperature to fall. A cool room helps; so does a hot shower a couple of hours beforehand, because it causes a subsequent drop.
NOISE AND SCHEDULE	Noise fragments sleep even when it does not fully wake you. And a stable schedule is worth more than any other measure in this table.

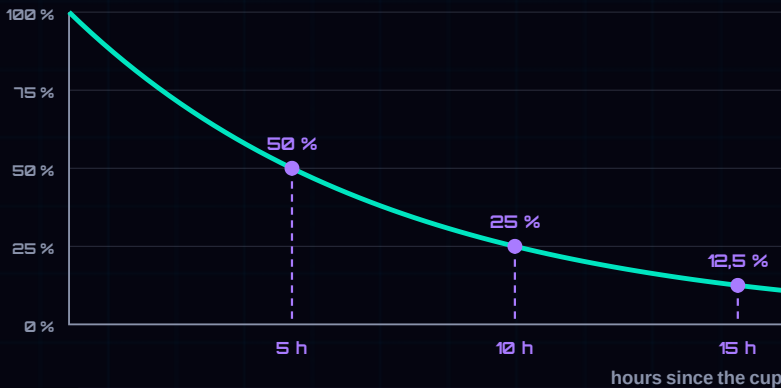
THE ONLY NON-NEGOTIABLE RULE ON THE LIST

The last hour without work and without screens showing activating content. Not because blue light is a poison —that headline is overblown— but because email, the news and arguments produce cognitive activation, and sleep does not arrive while the system is still solving something. You can read on a screen; you cannot work on one and fall asleep five minutes later.

CAFFEINE, ALCOHOL AND NICOTINE

Three legal substances, two of them consumed daily by half the population, and all three act on sleep in distinct and predictable ways. Caffeine deserves the bulk of the chapter because it is the most underestimated.

HALF-LIFE OF ABOUT 5 HOURS · BETWEEN 3 AND 7 DEPENDING ON THE PERSON



IT DOES NOT STOP YOU FALLING ASLEEP · IT CUTS THAT NIGHT'S DEEP SLEEP

THE CAFFEINE CLEARANCE CURVE Each half-life takes away half of what was left. With a half-life of about five hours, a quarter remains at ten hours and an eighth at fifteen. The figures are calculated from the half-life; they are not typed by hand.

Each half-life clears half of what was left. With five hours, a quarter remains at ten and an eighth at fifteen. The speed varies a great deal between people: some clear it twice as fast as their neighbour.

THE FOUR THINGS TO KNOW ABOUT CAFFEINE

IT DOES NOT
NECESSARILY STOP YOU
FALLING ASLEEP

It can let you sleep and still cut that night's deep sleep. That effect is not perceived, and hence the "it does not affect me".

THE SPEED VARIES
ENORMOUSLY

The half-life runs from about 3 to 7 hours among healthy people. Some clear it twice as fast as their neighbour.

IT TAKES TIME TO ACT

In the order of 20 to thirty minutes. That is what makes the coffee nap of the nap chapter possible.

IT ACCUMULATES

The five o'clock coffee adds to what is left of the eleven o'clock one. The day's total matters more than the last cup.

The test for whether it affects you is not to reason about it: it is to drop it for two weeks and compare the log. Any other answer is an opinion about yourself, and on this subject opinions about oneself are notoriously bad.

AND WITHDRAWAL, WHICH IS REAL

Stopping caffeine abruptly produces headache, irritability and fatigue for a few days in anyone who drinks it daily. It is not a sign of anything serious and it passes; it is worth knowing so as not to confuse

it with “quitting coffee disagrees with me”. Tapering down over a week avoids it almost entirely.



ALCOHOL: THE BEST-KEPT DECEPTION

It shortens the time it takes to fall asleep, and hence its reputation for helping. Afterwards it does the opposite: it fragments the second half of the night and suppresses part of the rapid eye movement sleep, which is exactly the sleep that lives there. The night starts easy and ends badly, with more awakenings, more thirst and a worse sense of rest.

A second effect matters to anyone who snores: alcohol relaxes the muscles of the upper airway and worsens apnoea episodes. Someone with undiagnosed apnoea has their worst nights precisely after drinking, and usually puts it down to something else.

NICOTINE

It is a stimulant, even though the act of smoking is experienced as relaxing. It lengthens the time it takes to fall asleep and, at high intake, produces early-morning awakenings as levels fall: a miniature withdrawal in the middle of the night. In the first days after quitting, sleep gets worse, and it is worth anticipating that so as not to read it as a reason to go back.

AND ONE ALMOST NOBODY LOOKS AT: MEDICATION

Several commonly used drugs disturb sleep: some antihypertensives, some antidepressants, corticosteroids, decongestants and bronchodilators, among others. If your sleep clearly got worse when you started a treatment, that is something to raise with whoever prescribed it. There is often an alternative, or a change in the time you take it, that solves it — and it is a two-minute conversation.

CHAPTER 7

THE BEDROOM

Sleep hygiene deals with behaviour; this chapter deals with the place. It is the cheapest part to fix and the one most people ignore while buying supplements.

THE FOUR PHYSICAL VARIABLES

TEMPERATURE	Cool. Falling asleep requires core temperature to drop, and a warm room prevents it. It is the variable with the greatest effect and the easiest to change.
LIGHT	Real darkness. Blinds that let the streetlight through, the standby LED on the television and the phone face-up on the bedside table all count. An eye mask costs little and works.
NOISE	Noise fragments sleep even when it does not wake you, and that partial awakening is not remembered in the morning. Earplugs, or a constant background noise that masks the peaks.
SURFACE	That it does not hurt. Beyond that, the mattress industry promises considerably more than it can demonstrate: the best pillow is the one that keeps your neck aligned in the position you actually sleep in, and that does not depend on the price.

THE RULE THAT TURNS THE BED INTO A CUE

This is the most important point in the chapter and it comes from the non-pharmacological treatment of insomnia, where it is called stimulus control. The idea is that the bed should be an unambiguous cue for sleep, and for that it cannot also be the place where you work, watch a series, argue, or turn over for an hour without sleeping.

- 1 The bed is for sleep and for sex. Nothing else: no laptop, no paperwork, no news.
- 2 Into bed only when you are genuinely sleepy, not merely tired.
- 3 If you have been awake for about twenty minutes and are starting to fret, get up and go to another room with dim light to do something boring. Come back when the sleepiness comes back. Yes, even if it is the third time.
- 4 No looking at the clock. Knowing the time does not help you sleep, and it does feed the mental countdown, which is what keeps you awake.
- 5 Get-up time does not change because of a bad night. It is the anchor, and moving it turns one bad night into a bad week.

These five rules look severe and they are the part that actually works. Getting out of bed at three in the morning is counter-intuitive, and it is precisely what breaks the association between the bed and frustration, which is what sustains insomnia once it has settled in.



SHARING A BED

Two people with different schedules, preferred temperatures and snoring in the same space is an engineering problem, not a romantic one. The solutions that work are unglamorous: separate duvets, earplugs, a mattress that transmits less movement and, when needed, separate rooms on some nights. Sleeping badly out of social habit has a real cost to the health of both.

PETS

An animal in the bed produces micro-awakenings that are not remembered and that show up as soon as you measure. If you sleep well, there is no reason to change anything. If you sleep badly and have an animal in the bed, it is one of the easiest variables to test for two weeks.

CHAPTER 8

THE NAP, DONE PROPERLY

The nap is a tool with instructions, and using it badly produces exactly the effect its detractors describe. With two well-chosen parameters—duration and timing—it does what it promises.

THE TWO DURATIONS THAT WORK

10 TO 20 MINUTES	You come out of light sleep, without grogginess, and alertness improves for a few hours. It is the default nap and the one that works for almost everything.
ABOUT 90 MINUTES	A full cycle, so you wake at the end of the cycle and not in the middle of deep sleep. It only makes sense with an accumulated deficit, or before a night you know will be short.
BETWEEN 30 AND 60 MINUTES	The bad window: a fair chance of waking in deep sleep, with that thick feeling that lasts half an hour.

THE HOUR

In the early afternoon there is a natural dip in alertness that is not caused by lunch—it happens without eating too—and that is the moment when a short nap fits at no cost. From mid-afternoon onwards it starts taking sleep pressure away from the night, and there it does compete with what matters.

The practical rule: a short nap, early, and only if you sleep well at night. Anyone with insomnia should avoid it, because part of the treatment for insomnia consists precisely of concentrating all the sleep pressure into the night.

THE COFFEE NAP, WHICH SOUNDS LIKE A GIMMICK AND IS NOT

Drinking a coffee immediately BEFORE a fifteen-minute nap works because caffeine takes on the order of twenty to thirty minutes to act: the nap happens before it arrives, and the two effects add up on waking. It is a practice that also appears in occupational fatigue-management settings, and it is the one thing in this book you can try today.

CHAPTER 9

WHEN SLEEP WILL NOT COME: WHAT ACTUALLY WORKS

This chapter deals with the insomnia that is not a diagnosed disorder but a bad patch, and describes the approach the evidence supports as first line — which is not a drug. If the situation is persistent, the final chapter says when to seek help, and you should.

THE LOOP THAT KEEPS IT GOING

Almost all insomnia starts from one cause —an upset, a change of schedule, an illness— and is then kept going by a different one: by what the person does to compensate for it. It is a recognisable loop:

HOW IT SETTLES IN

I SLEEP BADLY ONE NIGHT	Any cause at all. So far, normal
I COMPENSATE	I go to bed earlier, get up later, take a long nap, cancel things
I SPEND MORE TIME IN BED NOT SLEEPING	The bed comes to be associated with being awake and frustrated
I WORRY ABOUT NOT SLEEPING	Worry activates, and activation is the opposite of sleeping
I SLEEP WORSE	And back to the second step, now with more conviction

The key to the loop is in the second step, and it is counter-intuitive: compensating makes it worse. Spending ten hours in bed to get six hours of sleep produces sleep that is more fragmented and less deep than spending seven to get six and a half.

THE TWO LEVERS THAT WORK

STIMULUS CONTROL

The five rules from the bedroom chapter. The bed goes back to meaning sleep.

SLEEP RESTRICTION

Matching time in bed to what you actually sleep, and widening it little by little as sleep consolidates. It is the part that works best and sounds worst.

- 1** For one week, write down what time you go to bed, what time you get up and how much you think you slept.
- 2** Work out the average of actual sleep. Let us say it comes out at six hours.
- 3** Set a time-in-bed window of that figure plus half an hour, with a fixed get-up time. Six and a half, no more and no less.
- 4** When for a whole week you sleep most of that window, widen it by fifteen or twenty minutes. And so on.
- 5** Never go below a five-hour window, and do not do this on your own if you drive professionally, or if you have epilepsy, bipolar disorder or untreated apnoea. In those cases it is done with supervision.

The first nights you sleep worse and you are more tired during the day: that is expected and it is the mechanism, because what you are

doing is building up sleep pressure until the night consolidates. It usually starts to show in one or two weeks.



WHAT DOES NOT WORK, EVEN THOUGH IT LOOKS AS IF IT SHOULD

- Staying in bed “resting” without sleeping. It feeds the entire loop.
- Recovering with long naps. It consumes the sleep pressure you need at night. With insomnia, the nap is suspended.
- Going to bed much earlier. The wake maintenance zone guarantees you will not fall asleep, and it adds an hour of frustration in bed.
- Drinking to sleep. The chapter on caffeine and alcohol explains why it is precisely the worst idea on the list.
- Checking your phone “since I’m awake anyway”. Light, activation and the time on the screen, all three at once.

ON MEDICATION AND MELATONIN

Hypnotics have specific indications, short time frames and adverse effects, and that conversation is with whoever prescribes, not with a book. Melatonin is not a sleeping pill: it is a timing signal, and its usefulness depends on the dose and above all on the hour at which it is taken, which is what almost nobody takes into account. Taking it at bedtime “to sleep” is the most common use and the least supported.

CHAPTER 10

SHIFTS, TRAVEL AND BROKEN NIGHTS

Everything above assumes a life with stable hours. A large part of the population does not have one, and deserves a chapter rather than the useless advice to “keep a regular schedule”.

SHIFT WORK

Working nights forces you to sleep when the internal clock is pushing upwards, and to be awake when it is pushing downwards. There is no way for that to come free, and it is worth saying plainly because the literature is consistent: prolonged night work is associated with worse metabolic and cardiovascular health. What follows reduces the harm; it does not remove it.

WHAT HELPS

LIGHT DURING THE SHIFT

A well-lit environment in the first half of the shift helps maintain alertness.

DARK GLASSES ON THE WAY HOME

Leaving in daylight, the morning light tells the clock to wake up. Filtering it on the way home helps you sleep afterwards.

A COMPLETELY DARK ROOM

Here the eye mask and the blinds stop being a luxury.

CAFFEINE AT THE START

At the start of the shift, not at the end. The half-life arithmetic still holds even at night.

A NAP BEFORE THE SHIFT

A prophylactic nap before clocking on pays off more than trying to hold out and compensate afterwards.

FORWARD ROTATION

If the shift rotates, forwards —morning, afternoon, night— is tolerated better than backwards. It is easier to delay the clock than to advance it.



TRAVEL ACROSS TIME ZONES

Jet lag has one useful rule: travelling east means advancing the clock, and advancing it costs more than delaying it. That is why eastward journeys feel worse than westward ones across the same number of time zones.

- 1 Switch to destination time as soon as you board: your watch, your meals and your sleep.
- 2 On arrival, get out into daylight on the destination's schedule. Light is what sets the clock; everything else is accompaniment.

- 3 Going east, seek morning light at your destination and avoid it late in the day. Going west, the reverse: light in the afternoon and avoid it early.
- 4 Reckon on roughly one day per time zone to adjust fully. A three-day trip across six zones leaves time for nothing: in that case it is sometimes better NOT to adjust and to keep the home schedule.
- 5 In-flight alcohol makes all of the above worse, and cabin dehydration does not help.

THE BROKEN NIGHT NOBODY CHOOSES

A baby, an on-call shift, a sick relative. There is no optimising here, and there are three things that reduce the harm: sleeping in blocks even short ones rather than staying awake on continuous watch, sharing the nights with someone if possible, and protecting safety —not driving with accumulated sleep debt— above any other consideration.

THE RULE THAT IS NOT NEGOTIABLE

Sleepiness at the wheel degrades attention and reaction time comparably to other causes that are prosecuted by law, and unlike those it is not compensated for by willpower. If you are in a run of short sleep, this is the part of this book that can save your life: do not drive, or pull over and sleep for twenty minutes before going on.

CHAPTER 11

SLEEP AND TRAINING, IN BOTH DIRECTIONS

The relationship between the two runs both ways, and both directions are documented in public sources. It is worth treating them separately because the practical consequences differ.

WHAT EXERCISE DOES TO SLEEP

The physical activity guidelines list improved sleep among the benefits that appear after a single session, not after months. It is one of the best-described acute effects, and it is a useful argument for anyone who wants an immediate reward for training: the night after an active day is better than the night after a sedentary one, and you can check that the same night.

The known exception is very intense training very close to bedtime, which in some people delays sleep through activation and temperature. In others nothing happens. It is one of those things settled by testing for two weeks rather than by arguing.

WHAT SLEEP DOES TO TRAINING

WHAT DETERIORATES WHEN YOU SLEEP TOO LITTLE

RECOVERY	Tissue repairs itself above all during rest. A sustained deficit lengthens the time it takes to be ready for the next session.
PERFORMANCE	What falls most is the part that depends on coordination, attention and tolerance of effort, rather than raw maximal strength. It shows up sooner in a technical sport than in a squat.
APPETITE AND CHOICES	Sleep deficit is associated with more hunger and worse food choices. It is one of the reasons sleeping badly sabotages a diet without the person knowing why.
INJURY	Less attention and worse reaction time is exactly the scenario in which accidents happen in training, and outside it too.

The practical conclusion is uncomfortable for anyone who sleeps five hours and gets up early to train: in a week where sleep is very tight, trading a session for an extra hour of sleep is not laziness, it is a defensible decision. Tomorrow's session will exist; there is only one of tonight.



UNLOADING AND REST ARE NOT THE SAME THING

It is worth separating two things that the word “rest” runs together. When mechanical load is withdrawn from a body for weeks—bed rest, immobilisation, absence of gravity—it loses muscle, loses strength and

loses bone, in that order and at different rates. That is unloading, and it is deterioration.

THE FIVE EFFECTS OF UNLOADING, SO AS NOT TO CONFUSE THEM WITH RESTING

MUSCLE MASS	1 to 2 % per week
STRENGTH	Falls faster than mass does
BONE MINERAL DENSITY	Around 1 to 1.5 % a month
PLASMA VOLUME	Drops in the first few days
AEROBIC CAPACITY	Measurable within one or two weeks

The rest of this book is the opposite: it is the interval between loads during which adaptation happens. Without it, load produces not improvement but accumulation. A day without training inside a week that has training in it is rest; three months on the sofa are unloading. It is the same stillness with two opposite meanings, and it depends entirely on what surrounds it.

WHAT THE DOCTRINE SAYS ABOUT THIS

The U.S. Army's health and physical readiness manual treats sleep as one of its five domains of readiness, on the same footing as the physical and the nutritional, and describes management practices — schedules, planned naps, control of caffeine and light— for sustaining performance in conditions where sleeping well is not always possible. It is probably the most practical free document that exists on the subject, and almost nobody outside that institution has opened it.

CHAPTER 12

RECOVERY: WHAT MONEY CAN BUY AND WHAT IT CANNOT

The word recovery has become a commercial category: compression boots, cryotherapy, massage guns, drinks, devices. This chapter ranks all of it by what is known about each, and the ranking is uncomfortable for the industry.

IN ORDER OF REAL IMPORTANCE

1 • SLEEP ENOUGH	No argument and no substitute. It is the best-supported recovery intervention, the cheapest, and the least practised.
2 • EAT ENOUGH	Total energy and protein. A sustained deficit turns any programme into an accumulation of fatigue.
3 • PROGRAMME THE LOAD WELL	Lighter weeks, variation in intensity, an easy day after a hard one. Recovery is designed before you need it.
4 • MOVE GENTLY ON REST DAYS	Walking, swimming, easy cycling. Cheap, with a modest and real effect on how you arrive at the next day.
5 • EVERYTHING ELSE	Massage, cold, compression, devices. Small effects, mostly on how you feel, and none of them makes up for a failure in the first four.

THE CASE OF COLD, WHICH DESERVES ITS OWN PARAGRAPH

Getting into cold water after training reduces the sensation of soreness, and that is fairly well described. What is also described is that, used systematically right after strength training, it can blunt part of the adaptation that training was after: the inflammation being switched off is, in part, the signal.

The practical reading: if the goal is to perform in a competition tomorrow, cold makes sense. If the goal is to gain mass and strength over months, switching off the signal every day does not look like a good idea. It is a clean example of a tool that can be useful for one goal and counterproductive for another.



THE LIGHTER WEEK

Scheduling a reduced-load week every so often —four to eight weeks, depending on how hard the block is— is not wasting time: it is what allows the accumulated adaptation to show. You cut the volume roughly in half, keep some intensity so as not to lose your edge, and genuinely rest.

- The signs that it is needed sooner than planned: performance drops three sessions in a row, sleep gets worse, the desire to train disappears, and joint niggles pile up.

- Those four signs appear roughly in that order, and the first two are the ones you can see in the notebook.
- Backing off before you break costs a week. Backing off after you break costs months. That is the whole arithmetic of it.

THE DEVICES THAT MEASURE SLEEP

They estimate the stages with fairly limited reliability, and they produce a documented side effect: anxiety about the sleep score, which worsens precisely what is being measured. They are reasonably good for one thing —estimating total duration and seeing trends across weeks— and rather bad for the rest. If the morning number shapes your day, turn it off.

HOW TO CHECK IT YOURSELF

Here are the sources, named and with an indication of where to consult them, so that this book can be verified and corrected.

THE SOURCES OF THIS BOOK

DIETARY REFERENCE INTAKES	National Academies · distributed by the NIH Office of Dietary Supplements. RDA, AI and UL for vitamins and minerals.
PHYSICAL ACTIVITY GUIDELINES FOR AMERICANS, 2ND ED.	U.S. Department of Health and Human Services, 2018. Activity dose and the response curve.
DIETARY GUIDELINES FOR AMERICANS	USDA and HHS. Dietary patterns and limits for added sugars, saturated fat and sodium.
FOODDATA CENTRAL	USDA Agricultural Research Service. Food composition, searchable online at no cost.
FM 7-22 HOLISTIC HEALTH AND FITNESS	U.S. Army. Doctrine on physical training, nutrition, sleep and mental readiness, published in full.
THE NIOSH LIFTING EQUATION	National Institute for Occupational Safety and Health, CDC. Calculation of the recommended weight limit in manual handling.
RESEARCH ON UNLOADING AND BED REST	NASA Human Research Program. Effects of unloading on muscle, bone and the cardiovascular system.
REGISTER OF HEALTH CLAIMS	European Commission, with the scientific assessments of EFSA. What may be claimed about a nutrient, and what was applied for and refused.



THE ONES THAT SPECIFICALLY UNDERPIN THIS BOOK

CONCRETE ROUTES

THE SLEEP DOMAIN	FM 7-22, Holistic Health and Fitness, in the U.S. Army's publications library. Published in full. Look for the sleep readiness chapter: schedules, planned naps, caffeine and light.
RECOMMENDED DURATION	The public health recommendations on sleep duration in adults, published by the Centers for Disease Control and Prevention at cdc.gov, with their sleep hygiene section.
SLEEP AND EXERCISE	Physical Activity Guidelines for Americans, second edition, at health.gov. Improved sleep is listed among the acute benefits of a single session.
UNLOADING AND BED REST	NASA Human Research Program. Bed-rest studies, which are the source of the distinction between unloading and rest.

AND HOW THIS BOOK CHECKS ITSELF

The sleep figures on these pages are not written into the text: they live in a data module that the typesetting programme verifies before writing a single line. One of those checks is specific to this book:

- That 6 cycles of 90 minutes —9 hours— reach the 7-hour recommendation. If the cycle length or the number of cycles were changed to values that do not square with the recommendation, the book would not be written.
- That the four stages are complete with their four fields, which is what guarantees that the table in the text and the plate say the same thing.

- And the rest of the verified data module this book comes from: macronutrient ranges, fibre coherence, micronutrient ceilings, intensity equivalences and metabolic bands. If a single one of those checks fails, the generator stops and writes nothing.

The hour figures that appear in the first chapter are not typed in: they are calculated by multiplying cycles by duration. It is a small precaution with a concrete effect, namely that the text cannot age badly if the underlying figure is ever corrected.

THE REST NOTEBOOK

Four weeks. The first measures, the second anchors the schedule, the third removes what gets in the way and the fourth tests what has been learnt. None of them requires buying anything.

WEEK 1 · MEASURE WITHOUT CHANGING

- 1 Write down three things every day: what time you went to bed, what time you got up, and a one-word note on how you were in the afternoon.
- 2 Also note the time of the last coffee and the last alcohol, if there were any.
- 3 Do not use a sleep-tracking app. Its estimate of the stages is unreliable and it produces performance anxiety about something that gets worse precisely when you try.
- 4 At the end of the week, work out your average hours and your debt against the recommended 7. That number is the diagnosis.

WEEK 2 · SET THE ANCHOR

- 1 Choose a get-up time and hold it for all seven days, weekend included. All seven.
- 2 Get outside into daylight for ten or fifteen minutes within the first hour after getting up. Even if it is cloudy: the light of a grey day is

still far more intense than that of a room.

- 3 Do not touch your bedtime yet. It adjusts itself once the anchor is in place.
- 4 Note whether falling asleep gets easier by the end of the week. It usually shows around the fourth or fifth day.

WEEK 3 · REMOVE

- 1 Apply the countdown: caffeine out ten hours before, large meals and alcohol three, hard training two, work and activating screens one.
- 2 If you cannot manage all of them, apply just two: the caffeine and the last hour. They are the ones that pay off most.
- 3 Turn the lights down in the house during that last hour. Lamps instead of the ceiling light. It is a two-minute change with a real effect.
- 4 Note the night-time awakenings you remember. Compare with week 1.

WEEK 4 · THE BAD-NIGHT TEST

- 1 There will be a bad night. When it comes, do not compensate the next day by staying in bed longer or going to bed much earlier: hold the anchor.
- 2 You can use a short, early nap of ten to twenty minutes.
- 3 Note how long it takes you to get back to your average. With the anchor in place it is usually one night; without it, several.

- 4 Compare week 4 with week 1: average hours, awakenings and afternoon state. If all three improved without buying anything, the notebook did its job.

WHAT TO EXPECT

Most people who do these four weeks do not discover an exotic problem: they discover that they go to bed forty minutes later than they think, that the five o'clock coffee was not innocent, and that a fixed get-up time fixes more than all the night-time rituals put together. If in your case the four weeks improve nothing, that is valuable information too, and it is what to take to an appointment.

WHEN THIS IS NOT SLEEP HYGIENE

The closing chapter, and the limit. Sleep hygiene fixes the sleep of a healthy person with bad habits. It does not fix a disorder, and confusing the two makes people with a treatable problem spend years thinking they lack discipline.

SIGNS THAT YOU SHOULD CONSULT SOMEONE, NOT READ

- Loud snoring with breathing pauses that someone has witnessed, or waking up choking. With marked daytime sleepiness, headache on waking or blood pressure that is hard to control, it points to sleep apnoea, which is common, underdiagnosed and very treatable.
- Difficulty falling asleep or staying asleep at least three nights a week for more than three months, with daytime consequences. That is chronic insomnia, and it has a first-line treatment that is psychological, not pharmacological.
- An irresistible need to move your legs on going to bed, which improves when you move them. It has a name, identifiable causes — iron among them— and a treatment.
- Severe daytime sleepiness despite sleeping enough, irresistible sleep attacks, or a sudden loss of muscle tone with emotion.
- Acting dreams out physically, with sudden movements or shouting. It deserves assessment and is not a curiosity.

- Sleeping badly together with persistent low mood, loss of interest or continuous anxiety. There, sleep is a symptom, and treating it in isolation does not work.

None of these six is fixed with a cooler bedroom. All of them have an appointment as the first step, and most of them have a treatment that works well.



GLOSSARY

ANCHOR	Get-up time, which is the most powerful signal for setting the internal clock.
CYCLE	A complete pass through the sleep stages. About 90 minutes.
CHRONOTYPE	The individual tendency to function better in the morning or at night. It has a biological basis and is not changed at will.
SLEEP DEBT	The accumulated deficit against your need. It is only partly repaid.
HYPNOGRAM	A chart of the passage through the stages across the night. It is the first plate in this book.
N3	The one for physical recovery. If sleep is cut from the end, this one survives
SLEEP PRESSURE	The accumulated tendency to fall asleep, which grows with hours awake and is discharged by sleeping. It is what a long mid-afternoon nap consumes.
REM	Concentrated in the second half. If sleep is cut from the end, this is the one you lose
HALF-LIFE	The time the body takes to clear half of a substance. For caffeine, about five hours in an average adult.



And with that the book closes. It is built on public material almost nobody opens, with the figures verified before being printed and the

sources given by name and with their location, so that you can contradict it if you think it is wrong.

It promises no results and brings no secret method. It brings the architecture of a night, the arithmetic of what cutting it costs, a timetable for what to cut and when, the two levers that break the insomnia loop, and the list of signs that stop being this book's business and become an appointment's.

If only one thing from all these pages stays with you, let it be the fixed get-up time. It is the anchor of the whole clock, it costs nothing and it fixes more than any night-time ritual. With that, the decisions you make will be your own, which is the most an honest book can offer.

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